

Smart Post-ORIF Rehabilitation Ecosystem with Edge-AI Kinematic Tracking (AKIRA)

Hybrid IMU-Based Edge AI for Ankle Rehabilitation Exercise Recognition and
Postural Stability Assessment Following Ankle Fracture ORIF

J26-SE-304

PROJECT PROPOSAL REPORT

JAYASHAN S H S - IT23223608

**B.Sc. (Hons) Degree in Information Technology Specialized
in Software Engineering**

**Department of Software Engineering
Sri Lanka Institute of Information Technology**

September 2026

DECLARATION

I declare that this proposal is my own work and has been prepared for the individual research component titled “**Ankle Surgery Rehabilitation**” within the **AKIRA** group project. I have acknowledged all sources used and will verify the accuracy, validity, and originality of the material before submission. The proposed methods, research decisions, and interpretations remain my responsibility.

Student Name	Student ID	Signature
Jayashan S H S	IT23223608	

Supervisor Declaration

The above candidate is conducting research for their undergraduate dissertation under my supervision.

Supervisor Name	Date	Signature
Dr. Dinuka Wijendra	9/15/2026	

ABSTRACT

Ankle fractures are common musculoskeletal injuries and may require open reduction and internal fixation (ORIF), followed by a rehabilitation period in which restoration of ankle movement, functional control, and postural stability is important. Population-based evidence reports an ankle-fracture incidence of 168.7 per 100,000 people per year in one large decade-long cohort, while global burden estimates identify lower-leg fractures including ankle fractures among the most common and burdensome fracture categories [1], [2]. Conventional rehabilitation assessment relies substantially on clinical observation and periodic outcome measures, while wearable-sensor research shows promise for objective movement assessment but also reports methodological, validation, and clinical-translation limitations [3], [4], [5], [6].

This individual component investigates whether a two-IMU ankle sensing configuration can provide objective, repeatable measures of ankle kinematics and postural stability and use those measures to characterize longitudinal recovery after ankle-fracture surgery. The proposed component will synchronize and calibrate one IMU on the lower leg/tibia segment and one on the foot, estimate relative segment orientation, derive exercise and stability features, and combine repeated measurements into an Ankle Rehabilitation Progress Index (ARPI). Temporal sequence learning will be investigated for exercise recognition, while regression/forecasting model types will be evaluated for continuous recovery-state and clinician-defined milestone estimation. The research will follow a Design Science Research-oriented, iterative experimental approach with benchmark comparison, controlled validation, and usability assessment. The component forms the ankle-rehabilitation measurement and prediction layer of the complete AKIRA solution; it does not prescribe treatment or replace clinician decisions.

Keywords: ankle fracture ORIF; inertial measurement unit; ankle kinematics; postural stability; longitudinal rehabilitation

ACKNOWLEDGEMENT

I would like to acknowledge my supervisor, the research group members, and the academic and clinical stakeholders who will contribute to the development and validation of the Ankle Surgery Rehabilitation component. I also acknowledge the authors and researchers whose peer-reviewed work provides the evidence base for the proposed sensing, rehabilitation assessment, and validation approach.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
ARPI	Ankle Rehabilitation Progress Index
DOF	Degree of Freedom
F1	F1 Score
IMU	Inertial Measurement Unit
MAE	Mean Absolute Error
ORIF	Open Reduction and Internal Fixation
ROM	Range of Motion
RMSE	Root Mean Square Error
SUS	System Usability Scale

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1. Introduction

1.1. Background and Context

Ankle fractures are common musculoskeletal injuries that can require a prolonged rehabilitation period, particularly when surgical treatment is necessary. In cases where fracture alignment or stability requires surgical intervention, open reduction and internal fixation (ORIF) is used to restore and maintain the appropriate anatomical position of the fractured bones. A population-based study of 9,767 ankle fractures reported a mean annual incidence of 168.7 per 100,000 people, with falls and sports identified among the major injury mechanisms [1]. At the global level, the Global Burden of Disease analysis estimated 178 million new fractures in 2019 and identified lower-leg fractures, including ankle-related fractures, as an important contributor to the overall fracture burden [2]. These findings demonstrate the scale of ankle and lower-limb fracture recovery and support the investigation of practical technologies for monitoring rehabilitation.

Following ORIF, rehabilitation progresses according to the patient's clinical condition and recovery. Rehabilitation may involve gradual restoration of ankle movement, strengthening, balance, and functional activities, while progression is influenced by factors such as fixation stability, healing status, range of motion, strength, functional ability, and clinician assessment. Evidence from rehabilitation and postoperative weight-bearing research indicates that outcomes can vary according to clinical context, fixation stability, patient characteristics, and rehabilitation protocols [7] - [8]. Therefore, the proposed system is designed to support objective measurement and follow-up rather than establish a universal rehabilitation schedule. It will not prescribe exercises, modify weight-bearing restrictions, or override clinical instructions.

The specific problem addressed by this component is the difficulty of obtaining **low-cost, repeatable, and objective measurements of ankle exercise performance and postural stability across repeated rehabilitation sessions**. Clinical observation and functional assessment remain important, but assessments are generally performed at specific clinical visits and may provide only intermittent information about movement performance. Wearable inertial sensing provides an opportunity to capture movement data during rehabilitation activities and support repeated assessment outside conventional assessment intervals. However, previous research has identified important methodological considerations involving sensor placement, sensor-to-segment calibration, kinematic validity, and analytical assumptions [3] - [6]. These limitations indicate that a reliable sensing approach requires explicit calibration, synchronized measurements, and appropriate validation.

BEFORE



AFTER



Figure 1 -Real ankle radiograph showing the pre-operative fracture and post-operative internal fixation

For this component, two synchronized inertial measurement units (IMUs) will be positioned on the lower leg/tibia segment and the foot segment. The relative orientation between these two segments will be used to estimate ankle movement, with particular emphasis on dorsiflexion and plantarflexion. The proposed exercise set includes ankle pumps, ankle circles, ankle alphabet, dorsiflexion, plantarflexion, seated heel raises, standing heel raises, and single-leg balance. Movement-related measures such as ankle angle, range of motion, movement velocity, repetitions, trajectory characteristics, and movement variability will be considered together with postural-stability measures such as sway and stability-related motion variation. These measurements will be evaluated across repeated rehabilitation sessions and summarized using the proposed Ankle Rehabilitation Progress Index (ARPI) to represent changes in rehabilitation performance over time.

The proposed component therefore focuses on developing and evaluating a wearable-sensor-based approach **for** ankle exercise recognition, movement assessment, postural-stability assessment, and longitudinal rehabilitation monitoring following ankle-fracture ORIF. The component contributes to the wider AKIRA solution through its ankle-specific rehabilitation measurements while maintaining a clear responsibility boundary from the other group components. The research is aligned with Sustainable Development Goal 3 (Good Health and Well-Being) because it investigates technology-assisted monitoring that can support rehabilitation assessment and follow-up.

1.2.Literature Review

The literature review was conducted to address the practical and methodological questions that directly influence the proposed ankle-rehabilitation component. In particular, the review examined the suitability of wearable inertial measurement units (IMUs) for lower-limb exercise assessment, the factors that influence ankle and lower-limb kinematic estimation, approaches to sensor-to-segment calibration, and the validation requirements for wearable rehabilitation measurements. The selected studies were therefore considered based on their relevance to the methodological and design decisions of this research. The critical findings, strengths, limitations, and influence of each study on the proposed research are presented in Table 1.

Publication	Method / Dataset	Context	Key findings	Strengths	Limitations	Influence on this research
Elsoe et al. [1]	Population-based epidemiological study; 9,767 ankle-fracture patients over 2005–2014.	General ankle-fracture population; Denmark.	Mean incidence 168.7/100,000/year; falls 61% and sports 22%.	Large population and decade-long coverage provide strong epidemiological context.	Single regional population; does not evaluate digital rehabilitation monitoring.	Justifies the importance of an ankle-specific rehabilitation-monitoring problem and supports the need for objective follow-up.
Wu et al. [2]	Global systematic analysis using GBD 2019 data across 204 countries/territories.	Global fracture burden.	178 million new fractures in 2019; lower-leg/ankle category was among the most common and burdensome.	Very large global scope and standardized burden measures.	Not specific to ORIF rehabilitation or wearable sensing.	Supports the broader significance of fracture recovery technology.
O'Reilly et al. [3]	Systematic review of IMU-based lower-limb exercise detection/evaluation.	47 papers covering lower-limb exercise assessment.	Shows IMUs can support exercise detection and movement-quality assessment; clinical translation was limited in the reviewed evidence.	Broad synthesis of wearable exercise research.	Heterogeneous sensors, protocols, and validation; limited user/clinical evaluation.	Supports IMU use but motivates a validation-first, ankle-specific design.

Weygers et al. [4]	Methodological systematic review of IMU-based lower-limb joint kinematics validated against a reference system.	Studies using IMUs for lower-limb kinematic estimation.	Methods are application-dependent; biomechanical assumptions and prior information can affect estimates.	Explicit focus on methodological validity and reference-standard comparison.	No single universally optimal method; clinical implementation remains limited.	Motivates explicit calibration, application-specific kinematic definitions, and reference-based validation.
Pacher et al. [5]	Systematic review of lower-body sensor-to-segment calibration methods.	55 studies; manual, static, functional, and anatomical approaches.	Calibration is a major methodological issue and comparative evidence is still needed.	Directly addresses the sensor-to-segment problem.	Different evaluation references and methods make conclusions difficult.	Supports a dedicated synchronized calibration stage before kinematic analysis.
Prisco et al. [6]	Systematic review of wearable IMUs for gait analysis.	32 studies.	Good-to-moderate agreement for kinematics with optical systems; spatiotemporal agreement was [3]more variable.	Uses multiple validation approaches and summarizes agreement patterns.	Primarily gait-focused rather than post-ORIF exercise rehabilitation.	Supports error-based and agreement-based validation rather than relying only on classification scores.
Lin et al. [7]	Cochrane systematic review of rehabilitation interventions after ankle fracture.	31 studies of adults after ankle fracture.	Evidence was heterogeneous; some interventions affected activity limitation and adverse events.	High-level evidence synthesis and clinical outcome focus.	Heterogeneity prevented meta-analysis for many comparisons.	Supports measuring recovery objectively while avoiding assumptions of one universal rehabilitation protocol.
Rigney et al. [9]	Systematic review/meta-analysis of randomized controlled trials comparing early vs non-weight-bearing after ankle	6 RCTs; 1,190 patients.	Early weight-bearing improved short-term functional outcomes without increased complications in	High-level evidence and RCT-only inclusion.	Findings depend on stable fixation and patient selection; not a sensor study.	Reinforces the need for clinician-defined milestones and safety boundaries rather than autonomous treatment

	fracture fixation.		appropriately selected cases.			recommendations.
Semjonova et al. [8]	PRISMA-guided systematic review of functional foot/ankle biomechanical tests and wearable integration.	4 studies; 83 participants.	Wearable outcomes were feasible, but reliability, measurement error and MCID reporting were inconsistent.	Recent focused review combining biomechanics and psychometrics.	Very small evidence base and heterogeneous methods.	Strongly motivates repeatability testing, measurement-error reporting, and stakeholder validation in the proposed study.

Table 1 - Critical literature review

Critical Synthesis. The reviewed evidence supports the feasibility of wearable inertial sensing for lower-limb movement assessment but also demonstrates that reliable rehabilitation monitoring requires more than exercise classification. Sensor placement and calibration can directly influence kinematic validity, while rehabilitation outcomes may vary according to clinical condition, fixation stability, patient characteristics, and rehabilitation protocol [3]. Furthermore, the limited reporting of repeatability, measurement error, and clinically meaningful change in foot and ankle wearable-sensor studies highlights the need for systematic validation across repeated measurements [8]. Therefore, this research will investigate a calibrated two-IMU ankle-sensing approach that combines movement and postural-stability measures and evaluates their consistency across repeated rehabilitation sessions.

1.2.1. Existing Solutions and Their Limitations

For comparison, the existing approaches considered in this proposal fall into four broad groups: clinical visual assessment, laboratory motion capture, camera or depth-based exercise analysis, and wearable inertial sensing. Clinical assessment is accessible and clinically meaningful, but it usually provides measurements only during visits and can depend on the assessor. Motion-capture systems can offer detailed reference measurements, although they normally need specialized equipment and a controlled setting. Camera-based methods avoid body-worn sensors, but field of view, occlusion, lighting, and privacy can become issues. IMUs are more portable and simpler to deploy, but their accuracy depends strongly on placement, calibration, signal processing, and validation [3] - [6].

Commercial rehabilitation platforms and general activity wearables show that guided exercise and wearable monitoring are feasible. However, the evidence reviewed for

this proposal does not demonstrate a validated longitudinal system specifically for post-ORIF ankle rehabilitation that brings together two-segment ankle kinematics, exercise performance, postural stability, and clinician-defined recovery milestones. The aim here is therefore not to claim that the proposed system is better than a particular commercial product. Instead, it addresses the limitations identified in the reviewed research.

Existing approach	Typical function	Technology	Users	Key limitation relevant to this component
Clinical observation	Assess movement quality and function during visits	Clinician observation and clinical tests	Physiotherapists, orthopaedic teams, patients	Periodic, assessor-dependent, limited continuous quantitative data
Laboratory motion capture	High-precision movement measurement	Optical cameras and markers	Researchers, biomechanics laboratories	High cost, controlled environment, limited home use
Camera/depth exercise systems	Exercise detection and movement feedback	RGB/depth cameras and computer vision	Home-exercise users, clinicians	Occlusion/privacy/environment sensitivity; not always suitable for ankle-specific segment kinematics
Wearable IMU systems	Exercise detection and movement measurement	Accelerometers/gyroscopes with signal processing	Researchers, rehabilitation users	Calibration, placement, drift, validation and clinical-translation challenges

Table 2 - Existing Solutions and Their Limitations

1.3. Research Gap

The reviewed evidence demonstrates the potential of wearable IMUs for lower-limb movement assessment, while also identifying important limitations related to sensor calibration, kinematic validation, repeatability, and clinical translation [3] - [6]. Recent evidence specific to foot and ankle assessment remains relatively small and heterogeneous, with limited and inconsistent reporting of measurement error, reliability, and clinically meaningful change [8]. Therefore, the research gap addressed by this component is the lack of a validated ankle-specific longitudinal framework that combines two-IMU kinematic assessment with postural-stability measures to objectively describe rehabilitation progress over repeated sessions and estimate progress toward clinician-defined rehabilitation milestones following ankle-fracture ORIF.

2. Objective

2.1. Main Objective

The main objective is to design, implement, and evaluate a two-IMU ankle rehabilitation component that measures exercise kinematics and postural stability and then uses repeated measurements to describe recovery and estimate clinician-defined rehabilitation milestones after ankle-fracture ORIF.

2.2. Specific Objective

- **To develop a two-IMU sensing system** that captures synchronized lower-leg and foot motion data during prescribed ankle rehabilitation exercises.
- **To calculate and extract ankle kinematic and postural stability features**, including ankle range of motion, dorsiflexion, plantarflexion, movement consistency, and stability-related measures from the collected IMU data.
- **To develop and evaluate an AI-based assessment model** that identifies exercise performance and estimates the patient's rehabilitation status from the extracted movement and stability features.
- **To analyse repeated rehabilitation sessions and predict clinician-defined recovery milestones**, providing a longitudinal representation of the patient's rehabilitation progress following ankle-fracture ORIF.

3. Methodology

3.1. Research Methodology/ Research Design

This study will follow a **Design Science Research-oriented approach** supported by iterative experimental evaluation. The central research question for this component is whether two wearable IMUs can provide ankle movement and postural-stability measurements that are sufficiently accurate, repeatable, and interpretable to support longitudinal rehabilitation assessment following ankle-fracture ORIF.

The research therefore requires both the development and evaluation of a working artifact. The study will begin by defining the rehabilitation-monitoring problem based on the identified evidence and research gap. This will be followed by the design of the two-IMU sensing configuration, sensor-placement procedure, calibration process, and data-processing pipeline. Pilot data will then be collected to evaluate the feasibility and quality of the measurement procedure. The processing pipeline will be refined based on the pilot findings before exercise-recognition, movement-feature, and stability-feature evaluation.

The validated measurements will subsequently be used to construct session-level rehabilitation summaries and the proposed **Ankle Rehabilitation Progress Index (ARPI)**. Repeated-session measurements will then be analysed to represent changes in rehabilitation status and support estimation of progress toward predefined, clinician-defined rehabilitation milestones. Finally, the developed component will be evaluated using quantitative performance measures and usability feedback. This iterative process allows design decisions to be evaluated experimentally rather than assuming that a sensing configuration or analytical approach will be suitable without validation.

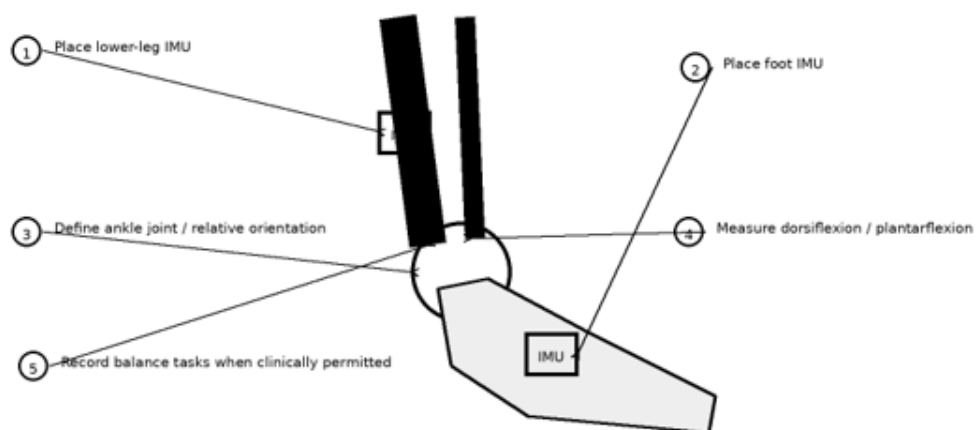


Figure 2 - anatomical reference showing the measurement procedure used in the ankle rehabilitation methodology

3.2.Data/ Participants/ Experimental Inputs

Three main sources of data are planned for this study. The first source will be **ankle exercise and balance data collected using the two-IMU prototype**. The second source will consist of **public or literature-derived non-clinical datasets**, which may be used for pipeline benchmarking when their sensor configuration, movement tasks, and data characteristics are sufficiently compatible with the proposed approach. The third source will be **longitudinal post-ORIF rehabilitation data** collected specifically for this research. Collection of post-ORIF data will begin only after the necessary institutional and ethical approvals have been obtained. Public datasets will not be treated as representative of post-ORIF rehabilitation unless their documented population, tasks, and context support such an interpretation.

For initial engineering validation, a pilot sample of approximately **20–30 adults** is being considered. A smaller healthy participant group may also be used during early protocol development and debugging of the sensing and processing pipeline. Participants with post-ORIF rehabilitation experience will be recruited only after ethical approval and will be assessed according to appropriate eligibility criteria, exercise restrictions, and rehabilitation stage. The final sample size will be determined based on feasibility findings, expected measurement variation, validation requirements, and guidance from the supervisor and relevant ethics process.

During each data-collection session, the two IMUs will operate using synchronized timestamps. Sensor placement, calibration information, participant identifier, and session identifier will be recorded before measurement. Accelerometer and gyroscope streams will be checked for missing or poor-quality samples and aligned temporally before further processing. Filtering and signal-processing parameters will be selected based on the characteristics identified during pilot testing rather than being fixed without empirical justification.

Exercise recordings will be labelled according to the performed exercise and, where applicable, repetition boundaries. The planned exercise set includes **ankle pumps, ankle circles, ankle alphabet, dorsiflexion, plantarflexion, seated heel raises, standing heel raises, and single-leg balance**. For balance activities, task duration and relevant trial outcomes will also be recorded where appropriate.

Data preparation will include:

- missing-sample and signal-quality checks;
- timestamp alignment between the two IMUs;
- sensor-offset and sensor-to-segment calibration;
- orientation estimation;
- movement segmentation;
- exercise and repetition labelling;
- feature extraction; and
- preparation of session-level rehabilitation measurements.

Normalization will only be applied where there is a clear methodological justification.

Training, validation, and test datasets will be separated **by participant**, ensuring that measurements from the same participant do not appear across different dataset splits. This

is intended to reduce the risk of participant-specific information influencing model evaluation.

3.3.Measurement and Processing Pipeline

The proposed measurement and processing pipeline begins with the placement of two synchronized IMUs on the **lower leg/tibia segment and foot segment** using a documented placement procedure. The sensor streams are synchronized before analysis, and calibration establishes the relationship between each sensor coordinate frame and its corresponding anatomical segment.

The accelerometer and gyroscope measurements will then be processed using sensor-fusion methods to estimate the orientation of each segment. The relative orientation between the lower-leg and foot segments will subsequently be used to obtain ankle-angle trajectories for the movement planes investigated in this study, with particular emphasis on dorsiflexion and plantarflexion.

The processing pipeline will contain two complementary analytical branches. The first branch will analyse dynamic movement data for **exercise segmentation, exercise recognition, repetition counting, and movement-quality assessment**. The second branch will analyse balance-task data for **postural-stability assessment**, including measures related to sway and sensor-motion variation.

The resulting features will be validated before being aggregated into session-level measurements. Where appropriate, measurements will be compared with the participant's baseline to represent individual change rather than relying only on population-level values. Repeated session summaries will then be used to calculate the proposed **ARPI**, which will provide a structured representation of rehabilitation progress. These longitudinal measurements will subsequently support recovery-state analysis and estimation of progress toward clinician-defined rehabilitation milestones.

The overall processing sequence is therefore:

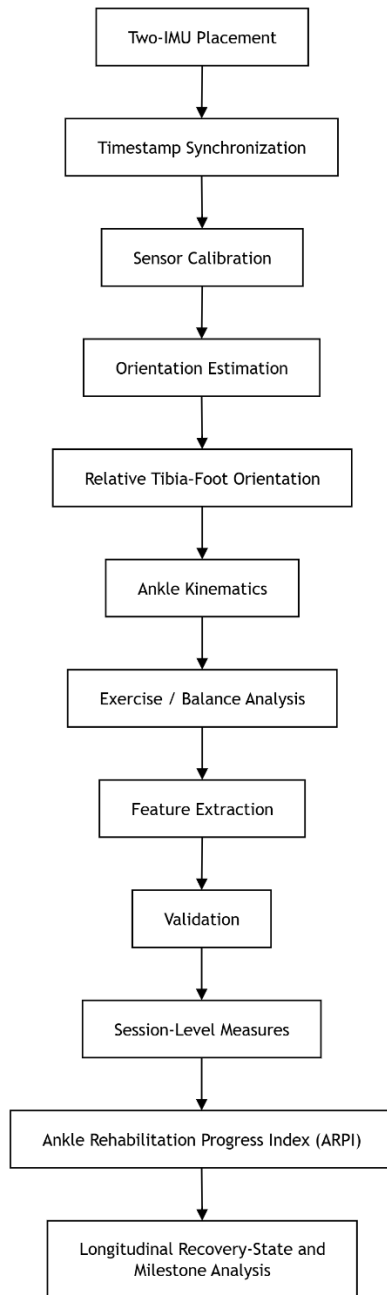


Figure 3 - Processing pipeline

3.4. Baseline/ Benchmark

For ankle-angle estimation, the primary comparison will be made against a **reference measurement procedure** selected during protocol finalization. Depending on the facilities available, this reference may be a clinician-based measurement procedure or laboratory-based reference kinematics. The purpose of this comparison is to determine whether the two-IMU approach provides sufficiently accurate ankle-motion measurements for the intended rehabilitation-monitoring application.

For exercise recognition, **simple signal-feature and rule-based approaches** will first be established as baseline methods. These baselines will provide a reference against which

the performance of the proposed temporal analysis approach can be evaluated. For recovery-state and longitudinal estimation, a **basic regression or trajectory-based approach using session-level rehabilitation features** will be used as the initial benchmark.

More complex analytical approaches will only be considered beneficial if they demonstrate measurable improvement over the established baselines under the same participant-independent evaluation procedure. This prevents model complexity from being treated as an advantage without corresponding evidence of improved performance.

The exact reference measurement procedure will be confirmed following the pilot feasibility assessment because access to a specific laboratory reference system cannot be assumed at the proposal stage. If a laboratory-based reference system is unavailable, the study will use a clearly documented reference protocol and explicitly report this limitation when interpreting the ankle-kinematic validation results

3.5.Evaluation Metrics

The proposed component will be evaluated using metrics that correspond to its main functions: exercise recognition, ankle-kinematic estimation, repetition analysis, postural-stability assessment, longitudinal recovery estimation, milestone prediction, and usability. The selected metrics and preliminary acceptance criteria are presented in Table 3.

Metric	Why Relevant	Target / Comparison
Exercise recognition Accuracy / Precision / Recall / F1	Measures correct classification while exposing class imbalance and false positives/negatives.	Target: $F1 \geq 0.90$ for the final exercise set, with per-class results reported.
Ankle-angle MAE / RMSE	Quantifies disagreement between estimated and reference kinematics in degrees.	Initial engineering target: $MAE \leq 5^\circ$ on the validated movement range; report confidence intervals.
ROM error	Directly measures error in clinically meaningful movement amplitude.	Target: median absolute ROM error $\leq 5^\circ$ in validated tasks.
Repetition-count error	Tests whether exercise completion can be quantified reliably.	Target: $\geq 95\%$ agreement with annotated repetition counts in controlled trials.
Stability measure repeatability	Tests whether the same task produces consistent measures across repeated trials/sessions.	Target: $ICC \geq 0.80$ where ICC is appropriate; report SEM/MDC when calculable.

Recovery-state MAE / RMSE	Measures continuous longitudinal estimation error.	Target: beat the simple baseline by $\geq 10\%$ on subject-independent test data.
Milestone time-to-threshold error	Measures usefulness of trajectory prediction for clinician-defined milestones.	Target: median absolute timing error ≤ 14 days in the final validated cohort, subject to sample size and milestone definition.
Usability / SUS	Measures whether intended users can operate the component effectively.	Target: SUS ≥ 70 after iterative refinement.

Table 3 - Proposed evaluation metrics and acceptance criteria

3.6. Validation Strategy

Validation will be conducted at multiple levels to assess both the technical performance and practical usefulness of the proposed ankle-rehabilitation component.

First, the **sensor placement, calibration, and ankle-angle estimation** procedures will be evaluated against the selected reference measurement procedure. This will determine whether the two-IMU configuration can produce sufficiently accurate ankle-kinematic measurements for the intended application.

Second, **exercise recognition and repetition analysis** will be evaluated using participant-independent data splits. Participants included in the test set will not appear in the training data, reducing the possibility that participant-specific movement patterns artificially improve the reported performance.

Third, **postural-stability features** will be evaluated for repeatability and agreement across repeated trials and, where data are available, repeated sessions. This will determine whether stability measurements remain sufficiently consistent to support longitudinal monitoring.

Fourth, **longitudinal recovery-state and milestone estimation** will use time-ordered evaluation. Later rehabilitation sessions will not be used to predict earlier sessions, thereby maintaining the temporal direction of the recovery-analysis problem.

The contribution of different feature groups will also be examined by comparing:

1. kinematic features alone;
2. postural stability features alone; and
3. combined kinematic and stability features represented through the proposed **ARPI**.

Where the available sample size permits, confidence intervals and error distributions will be reported alongside point estimates. Usability testing and stakeholder review will also be incorporated into the evaluation to identify practical issues that may not be captured by technical metrics.

Any clinical interpretation of the results will remain within the intended scope of the research. The system will be presented as a **rehabilitation-monitoring and decision-support technology**, rather than as a diagnostic system or an autonomous treatment system.

3.7. Development Approach

An **iterative development approach** will be used because several aspects of the proposed component—including sensor placement, calibration, signal processing, exercise analysis, and user interaction—may require refinement based on pilot findings. Each development iteration will produce a working version of the relevant component, evaluate it against the predefined criteria, document identified problems, and use the results to guide the next iteration.

The development process will proceed through the following stages:

1. **Prototype the sensing and synchronization layer**
Implement the two-IMU data-acquisition process and establish reliable timestamp synchronization between the lower-leg and foot sensors.
2. **Validate calibration and basic ankle-angle estimation**
Implement the sensor-to-segment calibration procedure and evaluate the resulting ankle-angle measurements against the selected reference procedure.
3. **Add exercise segmentation, recognition, and repetition analysis**
Develop the time-series processing required to identify the proposed ankle exercises, segment exercise repetitions, and calculate movement-related features.
4. **Add postural-stability analysis and ARPI calculation**
Implement stability-related feature extraction and combine validated movement and stability measures into the proposed Ankle Rehabilitation Progress Index.
5. **Evaluate longitudinal recovery-state and milestone estimation**
Use repeated session-level measurements to investigate rehabilitation-state changes and estimate progress toward clinician-defined milestones.
6. **Conduct usability and integration testing, then freeze the evaluated version**
Evaluate the component with intended users and relevant stakeholders, address identified usability and integration issues, and establish the final version used for the main evaluation.

This iterative sequence ensures that longitudinal analysis is performed only after the underlying sensing, calibration, movement, and stability measurements have been sufficiently evaluated.

4. High Level System Architecture

The full AKIRA solution is shown only at the level needed to explain integration. The individual contribution in this proposal is the Ankle Surgery Rehabilitation component, covering sensing, measurement, analysis, and longitudinal assessment. The detailed architecture and processing steps that follow therefore describe this component rather than the other group members' research areas.

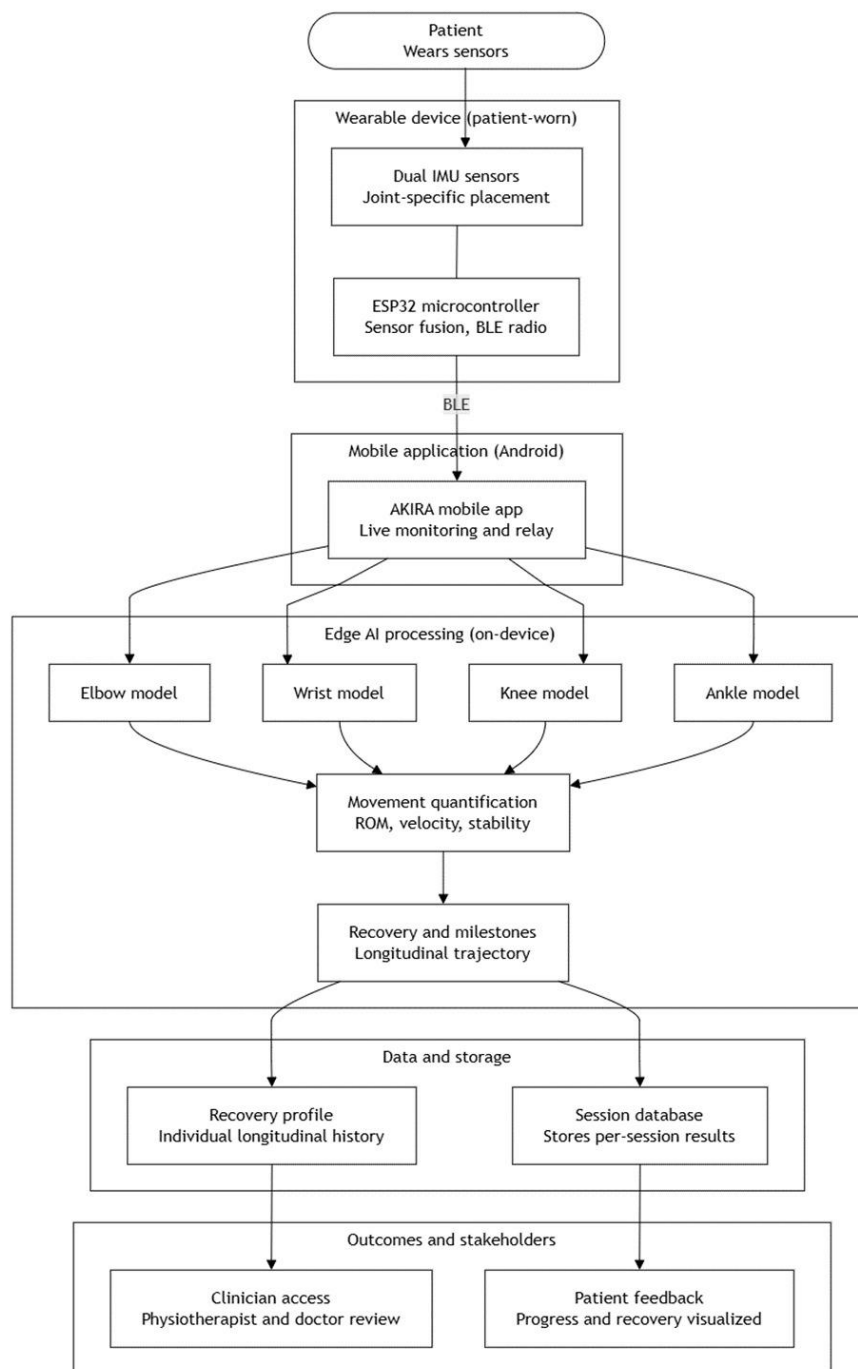


Figure 4 - High-level AKIRA Full Architect Diagram

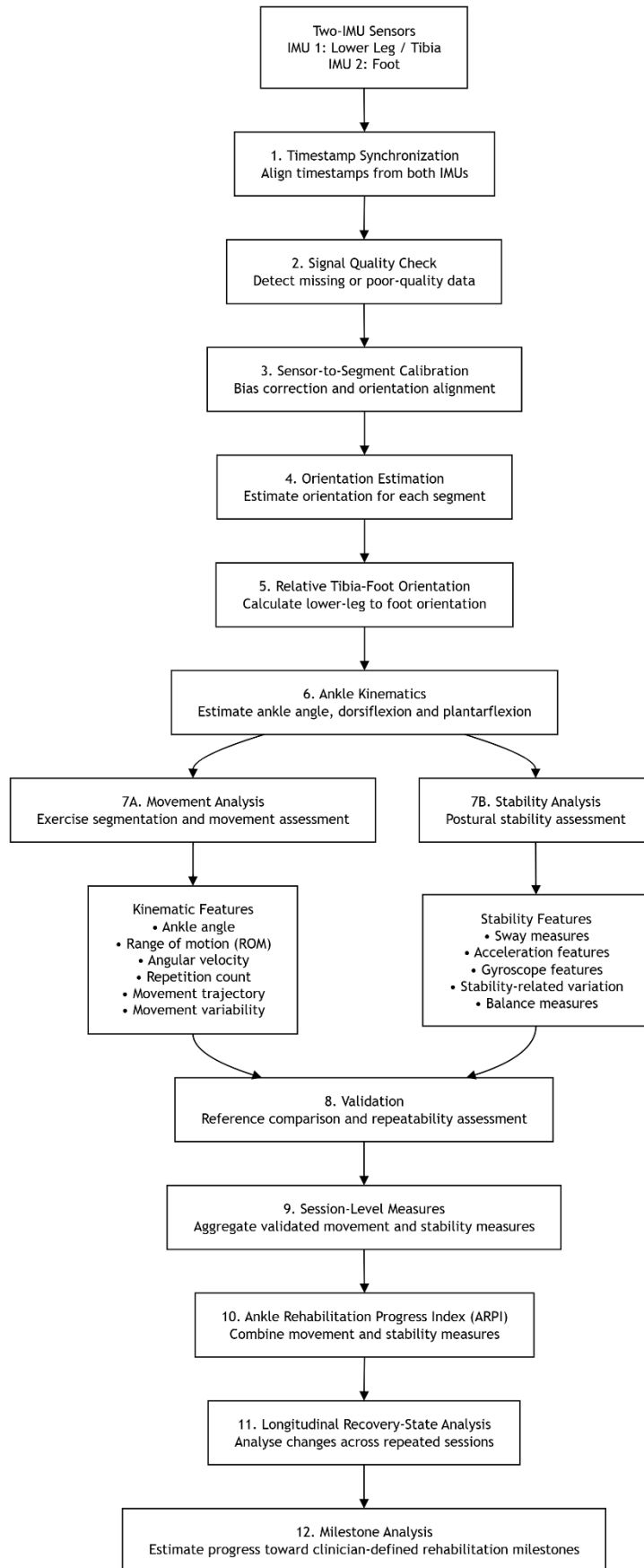


Figure 5 - Two-IMU ankle sensing and processing pipeline

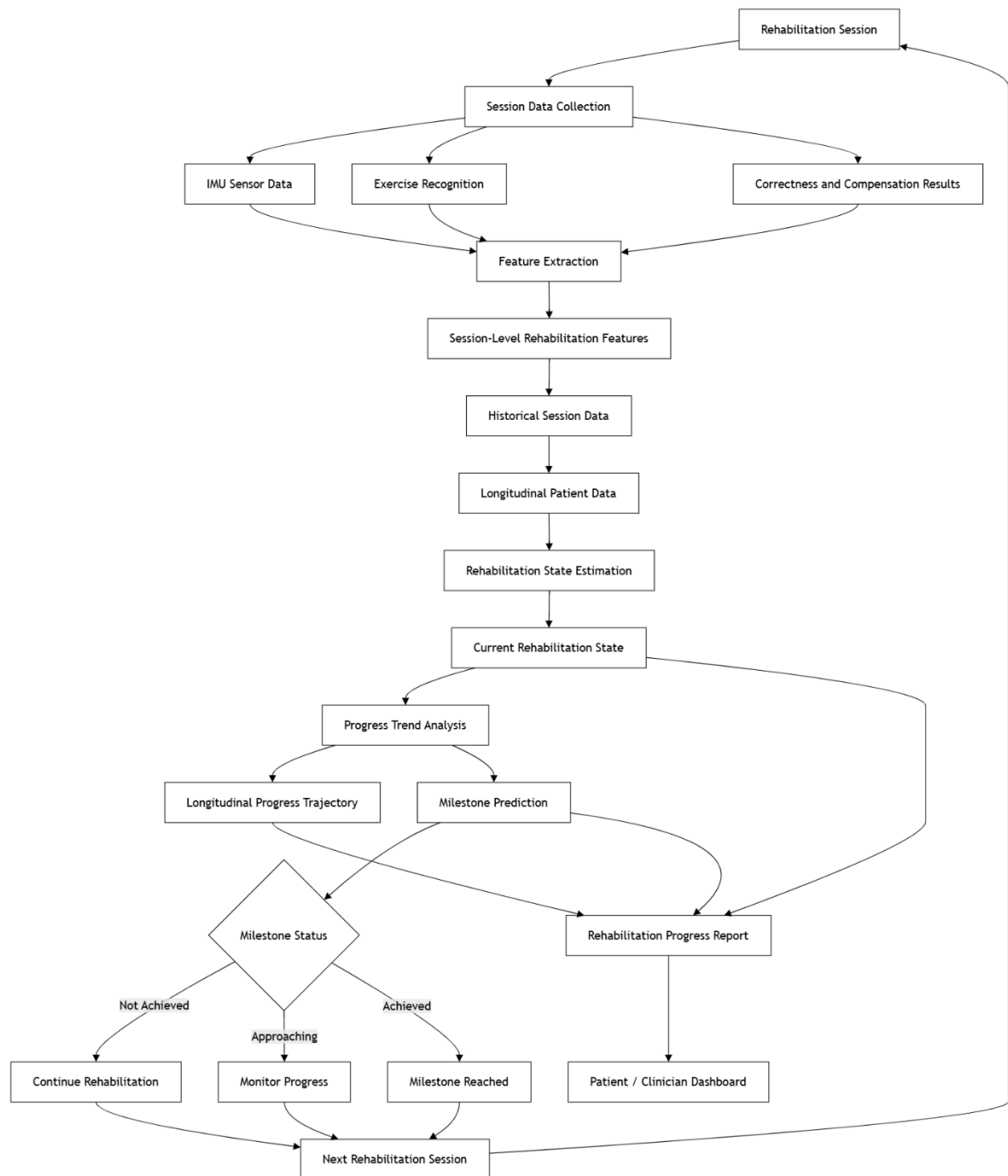


Figure 6 - Longitudinal rehabilitation-state and milestone prediction workflow

4.1.Overall Integration

The Ankle Surgery Rehabilitation component receives its own sensor/session inputs and produces structured ankle-rehabilitation measurements. At the group level, these outputs will be exchanged through the agreed project interface so that the complete solution can present a unified rehabilitation view. The exact API or data schema will be finalized jointly, but the individual responsibility boundary remains limited to ankle sensing, analysis, validation, and longitudinal outputs

4.2.Individual Responsibility Boundary

Activity / Deliverable	My Responsibility	Shared Responsibility	Other Member
Two-IMU ankle sensing and placement protocol	✓		
Ankle calibration and kinematic estimation	✓		
Exercise recognition and repetition analysis	✓		
Postural-stability analysis	✓		
ARPI and longitudinal milestone estimation	✓		
Integration interface / data exchange		✓	
Complete solution presentation		✓	
Other group components			Handled by respective members

Table 4 - Individual responsibility boundary

5. User Requirements

The requirements below are preliminary. They were derived from the research problem and from the intended rehabilitation workflow rather than from completed stakeholder interviews. The planned interviews, observations, or expert reviews should therefore be used to confirm or change these requirements before the final submission. The evidence gathered from stakeholders will be recorded and linked to the final requirement set.

5.1.Functional Requirements

- FR-01: The component shall acquire synchronized accelerometer and gyroscope data from two ankle-mounted IMUs.
- FR-02: The component shall support a documented calibration procedure for the tibia/lower-leg and foot sensors.
- FR-03: The component shall estimate ankle movement trajectories from relative segment orientation.
- FR-04: The component shall recognize the defined rehabilitation exercise classes using time-series sensor data.
- FR-05: The component shall calculate repetitions and exercise-level kinematic measures including ROM, angle, velocity, trajectory, acceleration, and variability.
- FR-06: The component shall calculate postural stability measures for supported balance tasks.
- FR-07: The component shall generate session-level summaries and the ARPI.
- FR-08: The component shall store session identifiers and timestamps so repeated measurements can be compared longitudinally.
- FR-09: The component shall estimate recovery-state trends and clinician-defined milestone timing when sufficient longitudinal data are available.
- FR-10: The component shall expose structured outputs for integration with the complete AKIRA solution.

5.2.Non-Functional Requirements

- NFR-01 **Accuracy**: final exercise-recognition performance shall meet or exceed the predefined F1 target on subject-independent test data.
- NFR-02 **Kinematic** validity: ankle-angle estimation shall report MAE/RMSE and remain within the predefined engineering error target on validated movements.
- NFR-03 **Reliability**: repeated trials shall report repeatability and measurement error rather than relying on a single accuracy result.
- NFR-04 **Usability**: representative users shall be able to complete the measurement workflow with a target SUS score of at least 70 after refinement.
- NFR-05 **Response time**: session-level analysis should complete within 5 seconds after the required sensor data are available on the target laptop/edge device.
- NFR-06 **Privacy**: identifiable participant information shall be separated from sensor records using coded participant identifiers.

- NFR-07 **Safety**: the component shall not independently prescribe exercises, modify weight-bearing restrictions, or override clinician instructions.
- NFR-08 **Reliability** of data capture: the system shall detect missing or invalid sensor samples and flag sessions that fail data-quality checks.
- NFR-09 **Reproducibility**: sensor placement, calibration, preprocessing, model version, and evaluation configuration shall be documented for every experiment.
- NFR-10 **Integration**: outputs shall use a documented machine-readable schema agreed with the group integration layer.

6. Commercialization Plan

6.1. Target Market and Customer Persona

The initial target market is physiotherapy and orthopaedic rehabilitation services that support patients after ankle-fracture surgery. Rehabilitation researchers and supervised home-rehabilitation programmes are possible secondary users.

The main user considered for the proposed system is a physiotherapist or rehabilitation professional who wants objective ankle movement and balance measurements between routine clinical assessments without setting up a laboratory motion-capture environment. From the patient side, the workflow is intended to remain simple and to operate within the rehabilitation plan provided by the clinician.

6.2. Value Proposition

- Portable two-IMU ankle measurement rather than laboratory-only motion capture.
- Exercise-specific quantitative measures instead of only subjective completion reporting.
- Separate kinematic and postural-stability measures to avoid reducing rehabilitation to a single movement score.
- Longitudinal ARPI and milestone trajectories that summarize change across sessions.
- Clinician-support positioning: monitoring and evidence generation without autonomous treatment decisions.

6.3. Revenue Model

One possible commercial model is a clinic package containing the wearable sensor kit, local analysis software, onboarding, and optional support or maintenance. A separate research or institutional licence could also be considered for universities and rehabilitation research groups. Pricing cannot be finalized at the proposal stage because it will depend on prototype costs, validation results, regulatory requirements, support needs, and feedback from potential users

6.4. Cost Estimation and Pricing Strategy

The budget shown in Section 7 should be treated as a development estimate for the research prototype, not as the final cost of a commercial product. A realistic selling price would need to include hardware, assembly, validation, support, replacement, and compliance costs. At this stage, the intended approach is to consider clinic value while keeping the hardware and software/support portions transparent rather than simply adding a margin to the student prototype budget.

6.5.Competitive Advantage

The main potential point of differentiation is the combination of ankle-specific two-segment kinematics, postural stability, exercise recognition, repeatability assessment, and longitudinal milestone tracking in one workflow. This remains a provisional research proposition. A stronger competitive claim would require a separate market and competitor study.

6.6.Intellectual Property Considerations

Possible intellectual-property outputs include the sensor-placement and calibration procedure, the feature-engineering approach, the ARPI formulation, the longitudinal analysis workflow, the software implementation, and the integration interfaces. Open-source libraries and datasets used during development will need to be checked for licence compatibility. Any decision about patentability, ownership, or public release should be discussed with SLIIT and the research supervisor.

7. Budget and Justification

Item	Estimated Cost (LKR)	Justification
Two IMU sensor modules / development units	20000	Core sensing hardware for the two-segment ankle configuration.
Mounting straps, fixtures and consumables	4000	Consistent and repeatable sensor placement.
Development accessories / cables / storage	3000	Prototype integration and data-transfer needs.
Printing / documentation / testing materials	2000	Protocol, consent and experimental documentation.
Participant session logistics	8000	Local travel/refreshments or session support subject to ethics and institutional rules.
Contingency	5000	Replacement components and unforeseen prototype expenses.

Table 5 - Preliminary component budget

Preliminary total: LKR 42,000. Software development will prioritize open-source tools where licensing permits. No cloud service is assumed as a required cost because the intended prototype analysis is designed to run locally/at the edge. The budget is an estimate and will be updated after hardware selection and ethics requirements are finalized.

8. Work Breakdown Structure

WBS ID	Task	Planned Timing
1	Literature review and evidence mapping	Weeks 1–3
2	Research design, ethics preparation and stakeholder requirement validation	Weeks 2–5
3	Two-IMU hardware setup and synchronization	Weeks 3–5
4	Calibration and kinematic pipeline	Weeks 4–7
5	Pilot data collection and preprocessing	Weeks 6–9
6	Exercise recognition and repetition analysis	Weeks 8–12
7	Postural-stability analysis	Weeks 9–12
8	ARPI construction and longitudinal analysis	Weeks 11–15
9	Model comparison and validation	Weeks 13–17
10	Usability and stakeholder evaluation	Weeks 16–18
11	Integration testing	Weeks 18–19
12	Final analysis and proposal/dissertation documentation	Weeks 19–22

Table 6 - Work breakdown structure

9. Grant Chart

The following high-level schedule aligns the individual ankle component with the required proposal activities. The exact academic calendar should be updated after confirmation of the official dissertation milestones.

Activity	Sep	Oct	Nov	Dec	Jan	Feb
Literature work	••	••				
Research design / ethics	••	••				
Data preparation / pilot		••	••			
Experiments			••	••		
Development		••	••	••	•	
Prototype work		•	••	••	•	
Testing / validation			•	••	••	
Integration				•	••	
Evaluation / analysis				•	••	•
Report preparation					••	••

Table 7 - Grant Chart

10. References

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11. APPENDICES

11.1. Appendix 1 — Mandatory Proposal Evidence and Decision Log

This appendix is intended to keep the proposal linked to the evidence and decisions made during the research process. It is left open so that new evidence, alternatives considered, and supervisor discussions can be added as the study progresses.

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion
Sep 2026	Wearable ankle monitoring requires validated calibration.	Literature review: [1] - [8].	Single IMU vs two IMUs.	Two-IMU tibia/foot configuration.	Two-IMU placement and calibration were discussed as necessary for reliable relative tibia-foot orientation and ankle kinematic estimation.
Sep 2026	Longitudinal recovery requires repeated, comparable measures.	Literature and protocol review.	Single-session score vs longitudinal index.	ARPI with repeated session measurements.	Longitudinal tracking using repeated session measurements and ARPI was discussed to represent rehabilitation progress over time.

Appendix 1 - Mandatory Proposal Evidence and Decision Log

11.2. Appendix 2 — AI Use Disclosure

Any generative AI assistance used during proposal preparation must be disclosed according to SLIIT requirements. AI assistance does not replace responsibility for factual accuracy, originality, reference verification, research decisions, methodology, technical choices, or interpretation of evidence. Technical diagrams required by the template will be created by the student and will not be generated by AI.

AI Tool	Purpose of Use	What Was Generated/Assisted	How Output Was Verified	Final Student Contribution
ChatGPT	Drafting / language formatting	Assisted with proposal prose and structure;	checked against supplied template, presentation, lecture guidance and verified literature metadata.	Research decisions, factual verification, originality, references, methodology, technical choices and final submission.

Appendix 2 - AI Use Disclosure

11.3. Appendix 3 — Preliminary Data Collection Protocol

- Record participant code, session code, date, and protocol version.
- Confirm clinician-defined exercise permissions and current rehabilitation stage before testing.
- Attach the lower leg/tibia IMU and foot IMU using the documented placement procedure.
- Perform the calibration routine and record sensor placement notes.
- Collect a short static/reference segment followed by the approved exercise sequence.
- Repeat selected exercises for reliability testing with rest periods as specified by the protocol.
- Collect balance-task trials only when clinically permitted.
- Store raw and processed data using coded identifiers and record any protocol deviations.

Appendix 3 - Preliminary Data Collection Protocol

11.4. Appendix 4 - Prototype Evidence

- Mobile App Prototype- [Figma Link](#)
- Web App Prototype – [One Drive Link](#)

Appendix 4 - Prototype Evidence

11.5. Appendix 5 - Questionnaire

Patients' Replies – [AKIRA Survey](#)

Appendix 5 - Questionnaire